

The Human Factor in Acquisitions: Cross-industry Labor Mobility and Corporate Diversification

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The benefits of internal labor markets are largest when they include industries that utilize similar worker skills, thereby facilitating cross-industry worker reallocation and collaboration. We show that diversifying acquisitions occur more frequently among industry pairs with higher human capital transferability. Such acquisitions result in larger labor productivity gains and are less often undone in subsequent divestitures. Moreover, acquirers retain more high-skill workers and more often transfer workers to jobs in other industries inside the merged firm. Overall, our results link human capital reallocation with the value created by corporate diversification and provide an explanation for seemingly unrelated acquisitions. (*JEL* G34, J24, J62, M51, M54)

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How important is human capital for determining the boundaries of the firm? In the presence of capital market frictions, a diversified firm can create value

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by reallocating capital internally from divisions with poor prospects toward divisions with better projects (Stein 1997). In a similar spirit, diversification can help firms to overcome labor market frictions. Internal labor markets create the opportunity to redeploy human capital to more productive uses in response to shocks while reducing the costs from search, training, or information asymmetry about worker quality. Moreover, they can increase productivity by enhancing worker and firm incentives for ongoing skill development and facilitating knowledge spillovers. Though increasing the scope of the firm may also come with costs, such as increasing influence costs or other types of agency conflicts within the firm (Milgrom 1988), the benefits of internal labor markets are most likely to dominate the costs when the component industries use worker skills that are more transferable.

We consider how the transferability of human capital affects the decision to diversify into new industries by acquisition. We find that economywide patterns of labor migration between industries significantly predict the industry choices made by acquirers in diversifying deals. Consistent with a human capital channel, high-skill workers are more likely to be retained and moved to new positions inside merged firms when the merging firms operate in industries that utilize similar skills. Moreover, mergers of firms with higher skill transferability are associated with larger increases in productivity than other diversifying deals.

To perform our analysis, we use data from the U.S. Census Bureau. We use the Longitudinal Business Database (LBD) to identify firms involved in acquisitions and the timing of transactions, and use data from the Longitudinal Employer-Household Dynamics (LEHD) program to link worker-level information to the acquiring and target firms and to measure patterns of movement in U.S. labor markets over time. We construct a measure of human capital transferability (HCT) using observed cross-industry worker job-to-job flows in the LEHD data. For each pair of industries in each year, we measure the frequency with which job changers move between the two industries as a fraction of the total number of job changers in the two industries over the prior 5 years. A relatively high fraction of job changers provides a credible proxy for the overlap in (or relatedness of) the skill sets required of workers in the two industries.

We then use our HCT measure to predict the choices of industries into which firms diversify through acquisitions. We regress the total employment (or number of firms) that is part of a deal in each industry-pair-year as a fraction of the total employment (or number of firms) that is part of any diversifying acquisition on our measure. We find that firms are more likely to diversify into industries that exhibit higher rates of job changes to and from the firm's industry in the external labor market. The relation is strongest when we measure the transferability of human capital using only the job changes of high-wage workers, whose human capital is likely to be scarce in the labor market. And, the result is robust to several approaches to control for product

market links between industries (i.e., input-output relations) and a variety of other factors that vary at the industry-pair level, such as asset, customer, or supplier overlap. Moreover, the relation between human capital transferability and deal frequency is strongest among industry pairs in which the benefits from labor redeployment are likely to be the largest because shocks are the least positively correlated.

Next, we examine the value implications of human capital transferability at the deal level. We show that, among diversifying acquisitions, deals in which there is high transferability of human capital between the acquirer and target perform better: the change in labor productivity in a 3-year window around the deal is about 18% higher. The results are strongest when we measure human capital transferability using only high-skill workers. The results also hold if we measure the change in labor productivity as a function of payroll instead of employment. Thus, firms that diversify across industries with high human capital transferability benefit even accounting for differences in wages they must pay to their workforces.

A challenge for the interpretation of this evidence is endogeneity of the HCT measure. Though it is backward-looking and defined at the industry-pair level, it is possible that it correlates with deal-specific factors that themselves cause improved postdeal performance. In our baseline analysis, we consider controls for potential omitted factors, such as supply chain linkages and asset overlap. As an additional step, we reestimate the relation using plausibly exogenous variation in the legal environments faced by acquiring firms to identify the effect of HCT on ex post productivity. Ex ante, firms in states that aggressively enforce noncompete agreements face more constraints on their ability to hire (skilled) human capital in the external market. Ex post, acquirers in those states have more power to retain workers inside the firm. Thus, the synergies from a deal to enrich the internal labor market should be higher where enforcement is strict. We confirm that acquirers in strict enforcement states experience higher relative productivity gains following high-HCT deals than acquirers in low enforcement states. Importantly, we do not exploit differences in the legal environment at the location of the target firm because target location is an endogenous choice. However, we confirm that the distance between the establishments of acquirer and target firms is significantly smaller among high-HCT diversifying deals than among low-HCT deals.

A natural question is whether the expected differences in productivity are reflected in higher announcement returns to high-HCT deals. To answer this question, we consider the subset of deals between publicly traded acquirers and targets using data from the Securities Data Company (SDC). We show that the combined cumulative abnormal returns are indeed significantly higher among high-HCT deals than among low-HCT deals, consistent with higher (perceived) deal synergies.

Next, we use worker-level data to provide direct evidence for the internal labor market mechanism. We examine worker retention inside the firm and migration within the firm to a different industry, four and eight quarters following a deal. We use these outcomes to proxy for the more extensive set of mechanisms through which internal labor markets improve productivity (knowledge transfers, increased worker investments in human capital, etc.), several of which are not directly observable. We find that there is a significantly higher retention rate of high-skill employees from the target firm following high-HCT diversifying deals, and that the merged firm is more likely to relocate workers from the target firm to a new industry previously operated by the acquiring firm. Because firms created in high-HCT deals also experience increases in labor productivity, our evidence suggests that the higher retention rates and redeployment create value.

As a final step, we link human capital transferability to the composition of established diversified firms. We find that industries between which workers migrate more freely in external labor markets are also more likely to be collocated in diversified firms, consistent with internal labor markets being a first-order explanation for firms' organizational structure.

Our results contribute most directly to the literature on the motives for corporate diversification. A prominent strand of the finance literature identifies corporate diversification as a symptom of agency problems, though the empirical link between diversification and value-destruction remains controversial.¹ Similarly, a number of papers analyze the differences in internal capital allocation between diversified and focused firms, but with mixed conclusions on the value implications (Stein 1997; Maksimovic and Phillips 2002; Villalonga 2004; Ozbas and Scharfstein 2010).² We instead analyze internal labor markets as a source of value creation in diversifying deals. We provide a new perspective on economies of scope as a source of synergies through the lens of human capital and internal labor markets. A literature in strategy, starting from Wernerfelt (1984), also takes a resource-based view of diversification. A common approach in that literature is to measure resource relatedness using overlap in the occupations employed in different industries (Farjoun 1994, 1998). The shortcoming of this approach is that there is no guarantee that workers sharing an occupation in different industries indeed find their skills transferable across the industries. We instead construct measures of cross-industry labor flows. Our approach is most similar to Neffke and Henning (2013), who link high cross-industry

¹ Papers linking diversification with agency problems include Betton, Eckbo, and Thorburn (2008), Morck, Shleifer, and Vishny (1990), and Silva (2016). Consistent with this notion, several studies find evidence of an empirical "diversification discount" (Lang and Stulz 1994; Berger and Ofek 1995), though some studies dispute the conclusion that diversification destroys value (Campa and Kedia 2002; Villalonga 2004; Arkan and Stulz 2016).

² See also Stein (2003) and Maksimovic and Phillips (2007) for a discussion of the literature on internal capital markets.

labor flows in Sweden with corporate diversification through greenfield investment.³ However, the internal labor market mechanism that we propose (two-directional dynamic labor reallocation, knowledge spillovers, etc.) is different from the mechanism that they investigate, which focuses on the redeployment of underutilized labor-in-place to new tasks. In a similar vein, [Beaumont, Hebert, and Lyonnet \(2021\)](#) show that high occupational overlap between new industries and firms' existing workforces predicts the choice to diversify by "building" rather than "buying." Distinct from these papers, we show that skill-relatedness predicts how firms expand their scope through acquisitions. We also provide novel evidence of the differential value implications of expanding into high- and low-relatedness industries, using differences in the preexisting legal environments facing acquiring firms as a source of plausibly exogenous variation in worker mobility, and provide direct evidence for the reallocation channel that we propose at the worker level.

We also contribute more generally to the growing literature linking labor economics to corporate finance. Many papers consider the role of labor markets in financing decisions ([Agrawal and Matsa 2013](#); [Simintzi, Vig, and Volpin 2015](#); [Graham et al. 2016](#)). Other papers link the mobility of human capital to the prices of risk in financial markets ([Donangelo 2014](#); [Eisfeldt and Papanikolaou 2013](#)). Unlike these papers, which model the firms from which human capital (may) exit, we consider variation in the transferability of human capital between industry pairs. By imposing this additional structure, we can link models of organization capital ([Atkeson and Kehoe 2005](#); [Lustig, Syverson, and van Nieuwerburgh 2011](#)) with the decision to expand a firm's scope through acquisitions. Our approach builds on the analysis of [Tate and Yang \(2015\)](#), who show that diversified firms use internal labor markets to transfer workers toward the industries with the best opportunities at a higher rate than workers make those moves in the external labor market. They also find evidence that workers develop skills inside conglomerates that facilitate such moves. However, their analysis takes organizational structure as given and, thus, cannot address whether such benefits are an economically important factor in the decision to diversify. Our results complement the analysis of [Ouimet and Zarutskie \(2020\)](#), who show that firms use takeover markets to expand the scale of their workforces, a mechanism that also can be particularly strong when firms need scarce skills or face constraints in external labor markets. We focus on investments that expand firm scope rather than scale. We do not argue that firms make acquisitions to obtain the services of particular workers, but instead to create internal labor market

³ [Neffke and Henning \(2013\)](#) use 280,000 job changes over a 4-year period in Sweden; our mobility index is constructed using 11.5 million external job changes in the United States over 15 years. Given the different application (M&A), we use a broader notion of industry (Fama-French 49 industry groups vs. four-digit European NACE codes, which are similar to four-digit NAICS codes), so that much of what they would classify as cross-industry jobs flows are intra-industry flows in our analysis. We compare and contrast the measures in the [Internet Appendix \(Table OA1\)](#).

synergies that exist regardless of the identities of the individual workers. Consistent with this distinction, we confirm that our results are strongest among industries that are subject to the least correlated shocks, a unique prediction of the redeployment mechanism. We also complement the findings of [Custodio and Metzger \(2014\)](#), who find that managers acquire assets in industries that they know, but who do not consider the transferability of human capital between acquirers and targets.

We also contribute to the large literature on the determinants of merger decisions. A number of recent papers focus on the effect of product market complementarities on acquisition decisions ([Ahern and Harford 2014](#); [Hoberg and Phillips 2012](#)). We focus on complementarities in input markets rather than output markets. [Rhodes-Kropf and Robinson \(2008\)](#) construct and test a merger model based on asset complementarities and search frictions. We extend the logic of their approach to study the implications of human capital complementarities. [Bena and Li \(2014\)](#) analyze technological similarities, measured using patent citations, as a source of merger synergies. Our results are complementary—we are likely to observe high human capital transferability (e.g., in R&D teams) where we observe technological overlap. However, we also find evidence that human capital transferability matters in industries (services) and among worker types (low skill) for which patents and technology are likely to be less important. Subsequent to our work, [Lee, Mauer, and Xu \(2018\)](#) show that occupational overlap between merging firms is associated with a reduction in employment following mergers, but has no effect on labor productivity, arguing for a labor redundancy mechanism that is distinct from our labor reallocation channel.

Finally, we contribute to the growing set of literature exploiting worker-firm matched data to study labor market mobility. Existing work finds strong patterns in the types of industry changes that workers make; for example, workers move more between industries with occupational overlap ([Bjelland et al. 2011](#); [Papageorgiou 2014](#); [Neffke, Otto, and Weyh 2017](#)). Moreover, geographic distance and the local industry mix interact with the choices that workers make ([Neffke, Otto, and Hidalgo 2018](#)). Another set of papers broadens the notion of mobility to include occupational as well as industry mobility (e.g., [Kambourov and Manovskii 2009](#); [Lagoa and Suleman 2016](#); [Groes, Kircher, and Manovskii 2015](#)). Given our focus on cross-industry corporate investment, we focus on cross-industry mobility; however, our analysis does not take a stand on the relative importance of the two sources of labor market mobility more generally. While these papers provide evidence of industry- or occupation-specific human capital, prior work also finds evidence that workers in diversified firms develop skills relevant to the portfolio of industries in which their firms operate ([Tate and Yang 2015](#)). Existing work also finds less industry or occupational mobility among high-skill workers, consistent with larger frictions that could increase the value of internal markets ([Parrado, Caner, and Wolff 2007](#); [Kambourov and Manovskii 2008](#)).

1. Theoretical Motivation

We motivate our analysis with a standard neoclassical merger model in which an acquiring firm purchases a target if and only if the combination produces positive synergies (i.e., a combined value exceeding the sum of the stand-alone values). We argue that the productivity gains from internal labor markets are a source of positive synergies in diversifying deals. The [Internet Appendix](#) provides a formal model to illustrate the gains from labor reallocation within an internal labor market. We do not argue that such gains are the only motivation for diversifying deals. For example, positive synergies could arise from asset complementarities ([Rhodes-Kropf and Robinson 2008](#)), or negative synergies from agency problems ([Morck, Shleifer, and Vishny 1990](#)). However, we develop our theoretical mechanism holding these other factors constant.

In our model, we assume that the adjustment costs firms face when transferring workers internally are less than the adjustment costs for external hires (for simplicity, we set them to zero). This wedge creates an advantage for a diversified firm in responding to an industry shock that causes changes to optimal labor input compared to a focused firm subject to the same shock. In response to a positive shock, for example, the focused firm must hire externally, paying the adjustment costs, while the diversified firm has the additional option to reallocate workers to the shocked industry from elsewhere in the firm. As long as it does not need to refill (all) the jobs of the transferred workers by hiring in the external market, this option produces a cost saving. The cost savings are highest when the industries in which the diversified firm operates simultaneously receive shocks in opposite directions.⁴ At the same time, transferring a worker to a new industry disrupts an optimal skill match so that transferred workers have a productivity discount compared to workers with optimal skills hired from outside the firm.⁵ The discount is smaller when the worker's existing skills are more transferable to the new industry. This transferability could arise because workers are employed in exactly the same occupations across industries (e.g., chemists can be employed in the chemical industry, as well as in the drug or food industries). Alternatively, it could

⁴ These savings can be large empirically. For example, we find that roughly 6.5% of workers from the target firm of a diversifying deal move to units of the acquiring firm in the first year after a deal. Assuming hiring costs equal to 24 weeks of pay ([Blatter, Muehlemann, and Schenker 2012; Abowd and Kramarz 2003](#)), an acquiring firm that acquires an average-sized target (300 workers) that pays average wages (\$31,545) would save roughly \$315,450 at the time of the deal without accounting for movement from the acquirer to the target or any dynamic benefits.

⁵ Note that because the worker's origin industry receives a negative shock at the time the firm wishes to redeploy her to a new industry, the worker's incentives are likely to align with those of the firm. The worker's job prospects in her home industry outside the firm are weak because other firms also will be cutting their workforces. Thus, the worker is more likely to accept redeployment inside the firm. More generally, internal redeployment allows workers to save on search costs and avoid external industry changes that come at significant wage costs ([Neal 1995](#)).

arise because distinct jobs in two industries nevertheless put positive weight on an overlapping set of general skills, in the spirit of Lazear (2009). For example, a physicist, computer scientist, and banker could all have analytical skills that would allow them to thrive at a hedge fund. Overall, the reallocation option in an internal labor market creates a net benefit to the firm when the adjustment cost savings exceed the productivity loss from skill mismatches, which is most likely when skills are transferable among the firm's industries and when the industries are more likely to experience simultaneous shocks in opposite directions. Thus, we make the following empirical predictions for firms' industry choices in diversifying deals:

Prediction 1. Diversifying mergers occur with a greater frequency between firms in industries between which there is higher transferability of human capital.

Prediction 2. The effect of human capital transferability on the frequency of diversifying deals is strongest among industries in which productivity shocks are less positively correlated.

In our framework, firms make acquisitions if and only if the expected synergies are positive. Because all actors have rational expectations, the average realized productivity changes following diversifying deals will increase with the transferability of skills across industries, generating our third empirical prediction:

Prediction 3. Among diversifying mergers, those in which the merging firms come from industries between which there is higher transferability of human capital achieve greater postmerger changes in productivity.

Our framework also generates an empirical prediction at the worker level. First, it directly follows that we should observe more internal movement of human capital postmerger when there is greater human capital transferability between the merging firms. The exact timing of this internal movement will depend on the realization of industry-level shocks over time. Second, we should observe more retention of high-skill workers following mergers between firms with greater human capital transferability. This effect should be immediately observable because it reflects optimal labor supply and demand choices based on the changes in incentives from the newly expanded internal labor market. On the demand side of the market, a broader internal labor market increases the firm's ability to retain high-skill workers in divisions hit by negative shocks by moving them elsewhere within the firm. This option is more valuable to the firm for high-skill workers because they are scarce in the external market so that hiring costs, including search, are likely to be higher than for low-skill workers.

On the supply side, workers should be more willing to stay in diversified firms because of the insurance provided by the internal labor market.⁶

Prediction 4. Among diversifying mergers, those that combine firms from high human capital transferability industry pairs are followed by more internal worker transfers and higher retention rates of high-skill workers.

In our formal framework, the economic mechanism that generates our empirical predictions is the relative advantage that diversified firms have in redeploying human capital in response to industry shocks. However, other economic mechanisms could reinforce these predictions. Worker productivity within a firm with a broader internal labor market could increase due to knowledge spillovers (Azoulay, Graff Zivin, and Wang 2010; Waldinger 2010; Jackson and Bruegmann 2009). For example, an engineer within General Electric's airline engine division could benefit from interacting with engineers in its wind turbine division in a way that might not occur if they were working in different firms due to lower coordination costs within the firm. In addition, workers could make larger investments in firm-specific human capital when the prospects of remaining in the firm are larger due to the enhanced job security provided by internal labor markets. Firms could formally encourage cross-industry investments via job rotation programs (Carmichael and MacLeod 1993). Consistent with these mechanisms, Boschma, Eriksson, and Lindgren (2009), Greenstone, Hornbeck, and Moretti (2010), and Timmermans and Boschma (2014) find that the flow of workers with related, but distinct skills into a plant (or county) increases plant productivity.

Finally, it is worth noting that our analysis focuses on the choice of industry conditional on the choice to expand the firm by diversifying acquisitions. We do not analyze the choice between expansion by acquisition or greenfield investment, both of which are methods to diversify the firm. Many factors that affect the choice between internal and external diversifying investment, such as time to build and investment scale, are likely to be orthogonal to our human capital transferability channel. Moreover, the internal labor market benefits that we study (dynamic relocation, productivity spillovers) are distinct from the one-time redeployment of underutilized human capital emphasized in the literature linking occupational overlap or skill relatedness with diversification by greenfield investment (Neffke and Henning 2013; Beaumont, Hebert, and Lyonnet 2021).

⁶ The existence of workers with transferable skills inside the internal labor market would make it cheaper for such firms to fire poorly performing workers (the firm could reallocate underutilized workers from elsewhere in the firm to save on some costs of hiring an external replacement). This in turn could make the firm less attractive ex ante to low-quality workers, which, if there is asymmetric information about worker quality, also could benefit the firm.

2. Data, Variable Definitions, and Summary Statistics

2.1 Construction of the sample and Human Capital Transferability Index

We use data from the U.S. Census Bureau's Longitudinal Business Database (LBD) and Longitudinal Employer-Household Dynamics (LEHD) program to test our hypotheses. The LBD covers all nonfarm establishments in the United States beginning in 1976 and contains information on plant ownership, location, status (active or inactive), industry, aggregate employment, and total payroll. We use the LBD to identify changes in ownership. We identify acquisitions as cases in which the firm identifier ("firmid") associated with all of a business's establishments changes from one year to the next. We also require that the new firmid existed in the data in the year prior to the change and that the old firmid transfers all of its operating assets to the new firm and disappears from the data following the change. The latter requirement eliminates partial asset sales from our data, but also reduces noise in merging the LBD to worker-level information in the LEHD data since all establishments in the firm (and, by implication, workers) experience the change. We use the SIC codes of all establishments of the merging firms to determine whether each ownership change is a diversifying transaction. We identify an acquisition as diversifying if there is no overlap in the Fama-French 49 industry codes of the acquiring and target firms' establishments prior to the deal. This definition of diversification excludes cases in which acquirers seek to expand their operations in the overlapping industries rather than to take advantage of human capital synergies across industries. Throughout the analysis, we define industries using the 49 Fama-French industries.⁷ We choose a broad definition of industries to maximize the distinction between the product markets in which firms in different industries operate and thereby to reduce measurement error.⁸ Our final sample includes about 4,000 diversifying acquisitions from 1995 to 2007.⁹ Because the LBD covers both public and private firms, we are able to analyze a substantially larger sample of deals than studies that focus on deals involving publicly traded firms.¹⁰ Moreover, there should be less

⁷ See Kenneth French's website (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) for industry definitions. We exclude the "Other" industry group. Our results do not depend on using this industry classification. Nevertheless, our measures of job flows will differ from those constructed by Hyatt et al. (2014) using NAICS classifications or Nefke and Henning (2013) using four-digit NACE classifications.

⁸ For example, upholstered household furniture manufacturing (337121) is a different six-digit NAICS industry from nonupholstered wood household furniture manufacturing (337122), both of which are different industries from wood office furniture manufacturing (337211). At this level of aggregation, firm classification is likely to be noisy: a firm that produces wooden chairs with cushions is also likely to produce wooden chairs without cushions.

⁹ Here, and elsewhere in the paper, we report sample sizes rounded to the nearest hundred, per Census policy at the time of our disclosure requests.

¹⁰ For example, Morck, Shleifer, and Vishny (1990) study 326 diversifying acquisitions; Kaplan and Weisbach (1992) study 123 diversifying acquisitions; and Chevalier (2000) studies a sample of 215 diversifying acquisitions.

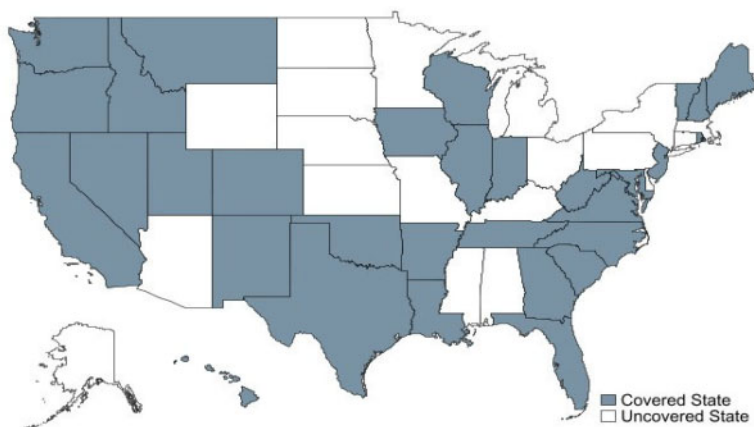


Figure 1
States covered in the Longitudinal Employer-Household Dynamics data

The map shows states for which worker-firm matched panel data are available through the U.S. Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) program in our sample. No data are available for states in white.

measurement error in our classification of diversification than in studies that classify deals using Compustat or CRSP because the LBD provides annual industry classifications for each of the firm's establishments and not just the firm's main industry code (Villalonga 2004).

Our worker-level information comes from the 2008 snapshot of LEHD data, which contain quarterly employer-employee matched observations for 31 states in the country. Figure 1 provides a map of the states from which data are available. There are no obvious regional biases in the set of included states, and we observe at least 42% of industry employment in all industries. Moreover, the reason for exclusion from the set of states available to researchers is typically preexisting state laws, suggesting that it is appropriate to consider the available states to be a random sample. The earliest available data come from 1990, though state entry dates vary. We observe workers' wages and the firm-unit in which they work as well as demographic traits including age, gender, and race.

We use the data for two purposes. First, we use the dynamic information on worker-firm matches to construct a measure of human capital transferability (HCT) across industries. We use worker job-to-job flows as a proxy for the broader transferability of workers' skills across industries. We construct an economywide index of worker mobility between Fama-French industry groups, which we refer to as the HCT index. Using a 10% random sample of the data, we identify about 11.5 million external job changes.¹¹ We observe the industry

¹¹ We use the Census "firmid" to identify firms so that we restrict our sample only to external job changers. We exclude moves to another reporting unit within diversified firms to limit endogeneity concerns when we link our

classification for each job at the reporting unit level.¹² We exclude workers with less than 1 year of tenure in their prejob change firms to limit the effect of temporary workers on our analysis. We also exclude workers in counties with fewer than 10 industries so that we are more likely to capture industry changes that reflect workers' preferences and not location constraints. By restricting the sample only to locations with broad industrial bases, this restriction also reduces the potential for differences in firm locations to confound our results, for example, from comparing firms in rural areas to firms in urban areas. Finally, we exclude large synchronized moves in which more than 10 workers from the same existing firm move to the same new firm during the same quarter because such worker flows might be themselves driven by ownership changes in the existing firm. For the remaining job changers in each industry, we trace the distribution of industries for the new jobs (including the original industry). We then aggregate across all quarters in a backward-looking 5-year window and scale by the total number of job changers in the industry. We use a 5-year window to minimize measurement error, which allows us both to observe sufficient numbers of job changes between all pairs of industries and to ensure that fundamental relationships are not overwhelmed by transitory shocks. For each pair of industries, we then compute a nondirectional measure of human capital transferability as the average of the fraction of workers from each industry that move to the other industry in the pair. It is important that our index is not directional because human capital transferability in either direction could motivate a deal. We also construct alternative versions of the measure in which we first restrict the sample (1) to workers earning at least \$75,000 in annual wages in their original jobs and (2) to workers earning less than \$25,000.¹³

Throughout the paper, we use industry fixed effects to correct for differences in the sizes of industries that could otherwise explain differences in the frequencies of cross-industry moves measured by the HCT index. The main purpose of the HCT index is to capture transitions among industries, not the macro conditions of the industries themselves. Thus, we compare the frequency of job changes from an origin industry to a particular destination industry only to the frequency of job changes to that same destination industry from different original industries and/or to other destination industries from the same original industry.¹⁴

mobility measure to diversification choices. Our use of a 10% random sample is innocuous and is necessary for tractability since the full LEHD sample contains hundreds of millions of observations.

¹² Reporting units are state employer identification numbers (SEINs). A single firm often reports many SEINs.

¹³ LEHD data reports quarterly wages. We annualize wages by taking the average quarterly wage over the prior four quarters and multiplying by four. We exclude the first and last quarters of workers' spells inside a firm to avoid bias from including wages earned over an unobserved portion of a quarter. We also adjust wages using the consumer price index so that the \$75,000 (\$25,000) threshold is consistent over time. The exact cutoffs do not drive our results.

¹⁴ Alternatively, we could incorporate information on differences in wage changes from different cross-industry moves. However, doing so could introduce noise because differences in wage changes are likely to differ

Our index uses worker industry changes in the external market to measure variation in the costs of internal redeployment from skill mismatch. We use external job changes because they can be measured more precisely. However, the frequency of external job changes across industry pairs could also vary with the costs of external hiring. In our model, we hold these costs fixed across industries. This assumption is reasonable for some sources of friction, such as asymmetric information about general ability. But, if instead costs are sufficiently large and positively correlated with skill transferability, they could reverse the correlation between skill transferability and job change frequency across industry pairs. In this case, we should observe the opposite patterns from the theoretical predictions in Section 1 when we use the HCT index to proxy for transferability. More plausibly, some of these costs, such as certain types of training costs, could negatively correlate with skill transferability. If so, the costs of postmerger labor adjustments in external markets could be lower for deals in high-HCT industries, regardless of the benefits from expanding the internal labor market. While this channel can also link industry choices to labor market frictions in a novel way, Prediction 2 provides a way to distinguish it from the internal labor markets channel. Moreover, it cannot by itself explain the choice to diversify because such costs will always be lowest for within-industry deals. In the [Internet Appendix \(Table OA5\)](#), we show that our baseline evidence is robust to using job changes within existing diversified firms to construct an alternative version of the HCT index, severing the direct link to external adjustment costs. We also consider other variations to the measure, such as splitting the index based on the direction of worker movement within an industry pair and directly adjusting for industry size.

We apply our measures of human capital transferability only within the universe of diversifying deals. While worker skills are likely to be the most similar across jobs in intra-industry deals, transfers of human capital do not generate the same type of benefits in focused firms that they do in diversified firms. For example, all workers are subject to the same industry shocks inside a focused firm so that the firm does not benefit from the ability to reallocate workers to divisions with the strongest investment opportunities. There may be many reasons why increases in scale can also generate synergies, and there is no a priori reason to expect the size of one type of synergy to dominate the other. Our focus is on the benefits of increases in scope via internal labor markets.

We also use the LEHD data to measure the consequences of mergers at the worker level. Because both the LBD and the LEHD data contain employer identification numbers (EINs), we can link firms in the LBD with quarterly worker information from the LEHD data for the firms involved in ownership

across voluntary and involuntary moves. As long as workers make optimal choices between possible destination industries, our measure will correctly capture differences in human capital transferability regardless of the motive for the job change.

changes in our sample. We use the longitudinal information in the LEHD data to identify job changes among workers in the target firms as well as movement to units owned by the acquirer prior to the deal. We exclude deals in which either the target or acquirer firm has establishments outside LEHD coverage (only) for our analysis of worker-level outcomes.

2.2 Summary statistics

Table 1 provides summary statistics for the data. In panel A, we provide details for the sample of job changers that we use to construct our measure of human capital transferability. The data include over 11 million worker-quarters, in which the average annual wage is \$24,990, and the average tenure in the prechange firm is roughly 3 years. In panel B, we provide summary statistics of the HCT index, the indexes defined within subsamples of high- and low-wage workers, and the industry-size-adjusted version of the index on the sample of distinct industry-pair years.

In panel C, we provide summary statistics of a subsample of merger deals for which we observe sales information for acquirer and target firms before the transaction and for the combined firm afterward and, thus, can measure the change in labor productivity. The information on sales is obtained from the Business Register (SSEL) and is only available for a subset of firms. Our subsample with sales information includes 3,900 deals, 600 of which are diversifying. Acquiring firms are much larger than target firms on average: mean employment among acquirers is 13,588; among targets it is 577. Labor productivity is similar among acquirers and targets, though it is significantly higher for acquirers than targets in the subset of diversifying deals, suggesting that diversifying acquirers operate relatively efficiently prior to the deal. We also observe that acquirers and targets in diversifying deals have smaller workforces than their counterparts in related deals. This difference is likely to come from defining diversification to include only deals without any industry overlap between the acquiring and target firms.

In panel D, we provide demographic information for the subsample we use for our worker-level analysis. The sample consists of 1.4 million workers from 3,700 transactions in which all involved establishments are in LEHD-covered states (roughly 400 of which are diversifying). In this subsample, the average worker is 39 years old, with 3 years of tenure in the target firm and predeal annual wages of \$30,350. Workers in target firms have higher mean wages than workers in the random sample in panel A, consistent with some firms targeting human capital when making acquisitions. Seventy-two percent of workers are white, and 41% are women. We also provide separate summary statistics for the subsamples of workers from targets in diversifying and within-industry deals. If anything, workers in the targets of diversifying deals appear to have higher wages (the difference is not significant), suggesting the importance of human capital in such deals. The only significant demographic difference is that we

Table 1
Summary Statistics

A. Sample of workers used to construct HCT Index

Tenure	%	Wage	%
1-2 yrs	39.7	<\$10K	24.2
2-3 yrs	20.9	\$10K – \$25K	42.4
3-4 yrs	12.7	\$25K – 50K	24.6
4-5 yrs	8.4	\$50K – 75K	5.5
>5 yrs	18.3	>\$75K	3.3
N (rounded)	11,521,900		
Mean	12.67	Mean	\$24,990
SD	9.87	SD	\$92,385

B. HCT Index (industry pair)

	Mean	SD
HCT Index	0.62%	1.01%
HCT (>\$75K)	0.56%	1.03%
HCT (<\$25K)	0.62%	1.07%
HCT_adj.	−0.30%	0.89%
N (rounded)	28,000	

C. Sample of acquisitions

	All	Related	Diversifying
Number of deals (rounded)	3,900	3,300	600
Acquirer's employment	13,588	14,877	6,638***
	(27,002)	(28,169)	(18,038)
Target's employment	577	629	296***
	(2,205)	(2,381)	(0,648)
Relative size	−2.81	−2.91	−2.25***
	(1.85)	(1.83)	(1.89)
Overlap	0.59	0.70	0
	(0.42)	(0.36)	
Acquirer: Sales/Emp	166	153	234***
	(267)	(245)	(354)
Target: Sales/Emp	171	170	176
	(220)	(219)	(225)
Acquirer: Sales/Payroll	4.11	3.90	5.27***
	(5.5)	(5.2)	(6.8)
Target: Sales/Payroll	4.70	4.76	4.39
	(5.1)	(5.1)	(5.0)

(Continued)

observe significantly more women in the targets of within-industry deals. We also observe higher worker retention rates following diversifying deals.¹⁵

Finally, in Table 2, we list pairs of industries between which we see a high frequency of worker movement using the industry-size-adjusted HCT index. Adjusting for the number of jobs in each industry allows us to interpret the values as deviations of observed mobility from what we would observe under random assignment of industry changers to new industries.¹⁶ In the

¹⁵ It is worth noting that the two subsamples (from panels C and D) do not completely overlap: one is restricted by the availability of sales information, and the other one is limited by the coverage of states in the LEHD data.

¹⁶ If we do not make this adjustment, industry pairs including Personal or Business Services, Wholesale and Retail, which are the largest industries, tend to rank highest. We present the top-10 rankings using the unadjusted index

Table 1
(Continued)

D. Workers in our acquisition sample

	All	Related	Diversifying
Number of deals	3,700	3,300	400
Tenure	12.18 (6.11)	12.18 (6.11)	12.13 (6.11)
Wage	30,350 (23,990)	30,201 (24,696)	31,545 (17,318)
Age	39.48 (4.74)	39.45 (4.78)	39.72 (4.37)
White	0.72 (0.21)	0.72 (0.21)	0.73 (0.22)
Female	0.41 (0.25)	0.42 (0.25)	0.37*** (0.22)
Retain($t=4$)	0.61 (0.21)	0.60 (0.22)	0.64*** (0.19)
Retain($t=8$)	0.45 (0.25)	0.45 (0.48)	0.48** (0.25)

This table presents summary statistics for samples used in the paper. Panel A describes the sample of job changers used to construct our measure of human capital transferability between industries (HCT Index). Panel B provides summary statistics for industry pairs. Panel C provides summary statistics of acquisitions by type, and Panel D provides summary statistics for workers employed by the target firms in acquisitions. The number of observations in summary statistics is rounded to comply with the Census's disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$ (difference in means between related and diversifying deals).

left columns of the table, we report the top-20 industry pairs when we use the full sample of job changes to measure the HCT index. To the right, we report the top-20 industry pairs when we measure the HCT index using only job changes among workers who earn annual wages exceeding \$75,000. We observe larger deviations from random assignment when we measure human capital transferability using the mobility of high-wage workers, consistent with the accumulation of skills that are not fully general, but more readily redeployed to specific subsets of industries. Using either measure, we observe some highly ranked pairs, such as Hardware and Chips, that also could be related through product market connections. To assess the overlap between human capital transferability and product market linkages, we use the Bureau of Economic Analysis' (BEA) input-output matrix to measure the fraction of industry output that flows between the two industries in the pair. We include a column in the table that indicates if the percentage exceeds 2.5. Using either HCT measure, we find that roughly 25% of highly ranked pairs share significant input-output links, suggesting that the two factors, while related, are distinct.

for comparison in [Internet Appendix Table OA2](#). In our regression analysis, we use the HCT index without adjustment; however, we include fixed effects for both industries within a pair, which implicitly makes the size adjustment.

Table 2
Summary of industry pairs based on the Human Capital Transferability Index

All workers							Workers with wage >\$75K						
Ind1	Ind2	Ind1_Des	Ind2_Des	HCT Index	Related (IO)		Ind1	Ind2	Ind1_Des	Ind2_Des	HCT Index	Related (IO)	
35	37	Hardw	Chips	3.43%	1		27	28	Gold	Mines	8.46%	0	
45	48	Banks	Fin	3.31%	0		45	48	Banks	Fin	4.86%	0	
20	21	FabPr	Mach	2.84%	0		35	37	Hardw	Chips	3.81%	1	
37	38	Chips	LabEq	2.53%	0		35	36	Hardw	Softw	3.59%	1	
35	36	Hardw	Softw	2.53%	1		24	26	Aero	Guns	3.43%	0	
12	13	MedEq	Drugs	2.43%	0		12	13	MedEq	Drugs	3.37%	0	
39	40	Paper	Boxes	2.36%	1		39	40	Paper	Boxes	3.20%	1	
24	26	Aero	Guns	2.24%	0		18	47	Chstr	RIEst	2.96%	0	
1	2	Agric	Food	2.23%	1		1	2	Agric	Food	2.83%	1	
17	21	BldMt	Mach	2.14%	0		34	36	BusSv	Softw	2.83%	0	
35	38	Hardw	LabEq	2.09%	0		2	3	Food	Soda	2.77%	0	
10	16	Clths	Txtls	2.00%	1		10	16	Clths	Txtls	2.75%	1	
27	28	Gold	Mines	1.92%	0		17	39	BldMt	Paper	2.74%	0	
21	23	Mach	Autos	1.89%	1		15	39	Rubbr	Paper	2.74%	0	
12	38	MedEq	LabEq	1.67%	0		37	38	Chips	LabEq	2.36%	0	
22	37	ElcEq	Chips	1.61%	0		15	40	Rubbr	Boxes	2.18%	0	
15	21	Rubbr	Mach	1.58%	0		19	20	Steel	FabPr	2.15%	0	
17	39	BldMt	Paper	1.48%	0		20	21	FabPr	Mach	2.06%	0	
21	22	Mach	ElcEq	1.45%	0		21	38	Mach	LabEq	1.99%	0	
17	20	BldMt	FabPr	1.43%	0		22	23	ElcEq	Autos	1.97%	0	
Mean				0.297%									
Std. Dev.				0.890%									

The HCT index measures the average rate (from 1990 to 2007) of movement across the two industries, adjusted for the number of jobs in each industry. We use the Fama-French 49 industry classification. We report the list based on versions of the HCT index using all workers in the left panel and workers with an annual wage higher than \$75,000 in the right panel. Related(IO) is an indicator variable that equals one if the percentage of industry output that flows between the pair of industries exceeds 2.5%, measured using the 1997 input/output matrix from the Bureau of Economic Analysis (BEA).

3. Human Capital Transferability and Diversifying Deal Frequency

We begin by testing Predictions 1 and 2, measuring the relation between the HCT index and acquisition frequency and how it varies with the correlation of shocks at the industry-pair level. Our goal in this section is to establish the consistency of industry-level patterns in diversification choices in the M&A market with the predictions from our model of internal labor markets. While we estimate associations that are robust to other known economic drivers of diversification decisions, the industry-level analysis does not allow us to make clear causal claims.

3.1 Prediction 1: HCT and industry choices in diversifying deals

Our model predicts that diversifying deals occur more frequently between firms in industries between which human capital transfers more easily (Prediction 1). To minimize the effects of firm-specific factors that affect both acquisition choices and the cross-industry transferability of human capital, we test the prediction at the industry level. We construct two dependent variables, both of which are measured annually at the industry-pair level. First, for each industry in which at least one acquiring firm operates, we aggregate the total employment acquired in all deals (both related and diversifying) in each distinct industry group. We then scale by the total employment across all deals within the acquirer's industry in that year. As such, the ratio we compute measures the absolute asset reallocation across industries and provides a fair comparison between industries. Second, we compute the same ratio using the number of firms instead of employment.

We then estimate the following regression specification:

$$Y_{ijt} = \alpha + \beta HCT_{ij(t-5,t-1)} + X'_{ijt} \gamma + \varepsilon_{ijt}, \quad (1)$$

where Y measures the intensity with which acquirers in industry i diversify into industry j (for $i \neq j$) in sample year t , HCT is the human capital transferability index between industries i and j measured over a 5-year window ending at $t-1$, and X is a vector of controls in year t .¹⁷ We include all of the 48 by 47 cross-industry pairs in the period between 1995 and 2007 for which we can define the HCT index (i.e., we exclude cases in which we do not observe any industry changers from either industry in the pair and, therefore, the denominator is zero).

To control for product market linkages across industries, we include in X a measure of the input/output flows between the acquiring and target industries, defined using the BEA's input-output matrix.¹⁸ Similar to the construction of

¹⁷ While our HCT index is symmetric between industry i and j , the dependent variable is not.

¹⁸ Here, we do not use a rolling 5-year window like we do to compute the HCT index because measurement error is less of a concern. Nevertheless, it is noteworthy that the rankings of industry pairs by input-output flows do not vary much from year-to-year over short horizons.

Table 3
HCT Index and diversifying acquisitions

Dependent variable	Pct_Acq(Emp)			Pct_Acq(NFirms)		
	All workers (1)	>\$75K (2)	<\$25K (3)	All workers (4)	>\$75K (5)	<\$25K (6)
A. All industry pairs						
HCT Index	0.432*** (0.064)	0.685*** (0.106)	0.274*** (0.056)	0.465*** (0.054)	0.703*** (0.100)	0.289*** (0.043)
Pct_Related	0.098*** (0.017)	0.050*** (0.019)	0.120*** (0.018)	0.108*** (0.015)	0.061*** (0.017)	0.132*** (0.016)
Acq. ind-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Target ind fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-square	0.06	0.07	0.06	0.10	0.11	0.09
N (rounded)	27,800	27,800	27,800	27,800	27,800	27,800
B. Excluding industry pairs with I/O relatedness > 2.5%						
HCT Index	0.440*** (0.071)	0.767*** (0.190)	0.308*** (0.052)	0.472*** (0.064)	0.796*** (0.187)	0.326*** (0.042)
Acq. ind-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Target ind fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-square	0.10	0.11	0.10	0.13	0.14	0.12
N (rounded)	24,000	24,000	24,000	24,000	24,000	24,000

This table reports coefficient estimates from OLS estimation of equation (1). Panel A includes all industry pairs, and Panel B includes industry pairs with an input/output relatedness less than 2.5%. In both panels, the dependent variable is the intensity with which acquirers in industry i diversify into industry j (for $i \neq j$) in year t . HCT Index is the human capital transferability index between industries i and j measured over a 5-year window ending at $t - 1$. The sample contains all industry pairs by year. The column header provides the sub(sample) of job changers we use to compute the HCT index. Columns 1 to 3 measure the frequency of diversifying acquisitions using the employment in target industries (Pct_Acq(Emp)), and columns 4 to 6 use instead the number of firms in target industries (Pct_Acq(NFirms)). Standard errors are clustered by the acquirer industry-year. The number of observations used in regressions is rounded to comply with the Census's disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

the HCT index (see Section 2), we compute the flows between two industries as the average of the flows in each direction. We include target industry fixed effects to control for time-invariant differences in the attractiveness of firms in different industries as takeover targets. Similarly, we include acquirer industry fixed effects to control for differences in acquisition propensities across industries. In our main reported specification, we allow these effects to vary by year to control for industry merger waves. The fixed effects also correct for differences in the sizes of industries that could contaminate the HCT index: we measure whether acquirers in a given industry are more likely to expand into industries with which they form high transferability pairs, adjusting for differences in the relative sizes of potential target industries. Our results are nearly the same if we also allow the target industry fixed effects to vary by year (see Internet Appendix Table OA3). Because industry choices by acquiring firms in an industry-year are not independent, we cluster standard errors by acquirer industry year.

We report the results in panel A of Table 3. In columns 1 to 3, we use the employment-based dependent variable to measure acquisition frequency. In column 1, we find that high human capital transferability between a

target industry and an acquirer's existing industries predicts significantly more diversification into that industry. Economically, a one-standard-deviation increase in the HCT index is associated with a 58% increase in the fraction of employment moving between the industry pair in diversifying deals from its mean, or 10% of a standard deviation. By comparison, a one-standard-deviation increase in industry relatedness measured by input-output links is associated with a 35% increase in the intensity of diversifying deals, or 5.5% of a standard deviation. Thus, human capital transferability explains more variation in firms' industry choices: the marginal effect of HCT is about 1.7 times the effect of product market links.

In column 2, we use the alternative version of the HCT index in which we measure human capital transferability using only job changes of workers who earn at least \$75,000 annually (high-skill HCT hereafter). The resultant estimate is economically larger. Here, a one-standard-deviation change in the index is associated with an increase in the intensity of diversification activity between the industry pair of 94% from its mean, or 15% of a standard deviation. In column 3, we use instead the index that defines human capital transferability using job changes of workers who earn less than \$25,000 annually (low-skill HCT hereafter). We find an estimate that is significantly positive, but economically smaller. This result could arise because of the greater availability of the skills of low-wage workers, a scenario that reduces the value of the redeployment option to the firm.

In columns 4 to 6, we repeat the same specifications, but using the dependent variable that measures acquisition frequency based on the number of firms involved in transactions. The employment-based measure better reflects the effect of the transactions on human capital allocation: deals involving more workers receive more weight. However, the results could conceivably be driven by a small number of very large deals. The firm-based measure instead equally weights transactions. Despite the differences, we find remarkably similar estimates of the relation between human capital transferability and diversification. Again, firms are significantly more likely to diversify into industries that have skill overlap with their existing portfolios of industries. The association is largely driven by high-wage workers and the effects are economically stronger than the effect of input-output relations between industry pairs. In panel B, we implement a less parametric control for input-output relations, dropping any industry pair in which more than 2.5% of output flows between the two industries. The results are essentially unchanged.

In [Internet Appendix Tables OA4–OA7](#), we report a number of additional robustness checks. We confirm that the relation between HCT and industry choices in diversifying deals continues to hold with direct controls for physical capital transferability (captured by asset overlap), customer or supplier overlap, occupation overlap, and other industry-pair differences that could correlate with the HCT index (industry growth rates or Tobin's q , average firm size or age, average wage, etc.). The high-skill HCT result continues to hold if we

include industry-pair fixed effects to capture any time-invariant unobservable at the industry-pair level, despite the fact that much of the meaningful variation in HCT is in the cross-section. We also show robustness to a variety of changes to the construction of the HCT index (size-adjustment, a 1-year look-back window, internal instead of external job changes, different ways of treating the direction of worker movement).

3.2 Prediction 2: Correlation in growth opportunities

Our model predicts larger net benefits from reallocating labor internally instead of tapping external markets at times when optimal labor adjustment differs in direction across divisions (Prediction 2). In these situations, workers from the division with bad opportunities who are transferred to the division with good opportunities do not need to be replaced in their original jobs, preserving maximal adjustment cost savings for the firm. Dynamically, this scenario is more likely if the industries that define the divisions experience less correlated shocks.

To test this prediction, we measure the correlation of employment between each pair of industries over the 15-year period just preceding our main sample (1980 to 1994) and use it to split industry pairs into quartiles.¹⁹ We observe a wide distribution of correlation coefficients across the full set of industry pairs (mean = 0.07; SD = 0.59). Unsurprisingly, we observe somewhat higher correlations among the top-20 high-HCT pairs from Table 2 (mean = 0.15; SD = 0.56); however, the distribution remains wide and includes some pairs with negative correlations.²⁰ We illustrate the distributions among the high-HCT and remaining pairs in Figure 2.

In Table 4, we report the results from estimating Equation (1), but allowing the coefficient for the HCT index to differ for industry pairs in the top and bottom quartiles of employment correlation (the level effect of HCT measures the effect for firms in the middle-two quartiles). We find that the HCT index has a significantly stronger association with the frequency of diversifying deals among industry pairs with the least correlated employment shocks. In column 2, we repeat the estimation using the high-skill HCT index. We find similar results. Thus, we confirm that the positive association between the HCT index and diversifying deal frequency comes from the set of industry pairs predicted by the theory; that is, industry pairs in which changes in optimal labor inputs are less correlated, creating larger expected benefits from internal reallocation.

4. Human Capital Transferability and Acquisition Performance

Our model also makes a prediction for postdeal performance: firms will experience greater gains in productivity following acquisitions that diversify

¹⁹ Results are qualitatively similar when we estimate correlation over a 5- or 10-year period.

²⁰ These summary statistics are calculated using Compustat data to avoid the need for a disclosure review.

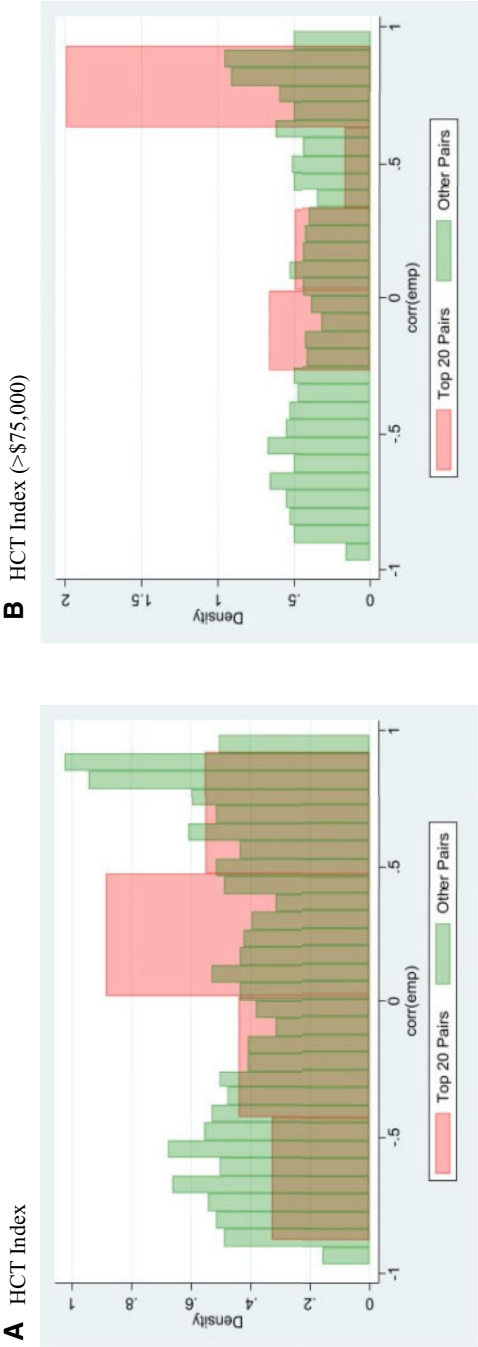


Figure 2
Industry pairwise correlation by HCT index

The map shows the distribution of industry pairwise correlation by HCT index group. The top-20 pairs are selected based on HCT Index (panel A) or HCT Index (>\$75,000) listed in Table 2 (panel B). The pairwise correlations are estimated based on industry total employment from 1980 to 1994.

Table 4
HCT Index and diversifying acquisitions: Correlation in growth opportunities

Dependent variable	Pct_Acq (Emp)	
	HCT using all workers (1)	HCT using high-skill workers (2)
HCT Index	0.556*** (0.075)	0.422*** (0.064)
High_Corr	0.004*** (0.001)	0.001 (0.001)
HCT × High_Corr	−0.231*** (0.083)	0.231** (0.100)
Low_Corr	−0.001 (0.001)	−0.002 (0.001)
HCT × Low_Corr	0.945** (0.419)	0.888** (0.449)
Pct_Related	0.087*** (0.017)	0.054*** (0.017)
Acq. ind-year fixed effects	Yes	Yes
Target ind fixed effects	Yes	Yes
t-stat for the Difference	3.89***	2.02**
Adjusted R-square	.06	.07
N (rounded)	27,800	27,800

This table reports coefficient estimates from OLS regression at the industry pair level of the frequency of diversifying deals on the HCT Index, the correlation of shocks, and their interaction. The dependent variable is the intensity with which acquirers in industry i diversify into industry j (for $i \neq j$) in year t measured by the employment of target firms (Pct_Acq(Emp)). HCT Index is the human capital transferability index between industry i and j measured over a 5-year window ending at $t - 1$. Column 1 uses the HCT index constructed using all workers in the sample, and column 2 uses the index based on workers with an annual wage greater than \$75K. High_Corr (Low_Corr) is an indicator variable that equals one if the correlation of employment between two industries (from 1980 to 1994) is in the top (bottom) quartile among all industry pairs. t-stat is shown for testing the difference between (HCT × High_Corr) and (HCT × Low_Corr). Standard errors are clustered by the acquirer industry-year. The number of observations used in regressions is rounded to comply with the Census's disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

the firm into industries with human capital that is more transferable to the firm's existing industries (Prediction 3). This prediction is central to the policy implications of our analysis, yet a simple correlation between deal performance and the HCT index is challenging to interpret due to endogeneity concerns. For example, high-potential industries could attract both corporate investment in the form of acquisitions and workers looking to make career changes. In this section, we test the prediction at the deal level, considering several different measures of productivity gains and several deal samples. Exploiting the deal-level setting, we also use plausibly exogenous cross-sectional variation in the legal environments facing different acquiring firms to try to isolate a causal channel.

4.1 Prediction 3: HCT and postdeal changes in labor productivity

We measure labor productivity using the ratio of firm sales to employment, as in, for example, Haltiwanger, Lane, and Spletzer (1999), Tate and Yang (2015), and Giroud and Mueller (2011).²¹ To account for wage changes around

²¹ Since we include both manufacturing and nonmanufacturing firms (units) in our sample, we do not have enough information to calculate total factor productivity. However, Foster, Haltiwanger, and Krizan (1998) show that

acquisitions, we also consider the ratio of firm sales to payroll. We construct our measures of labor productivity using firm-level sales information from the Census Bureau's Business Register (SSEL), and information on employment and payroll from the LBD. We compute changes in labor productivity around acquisitions over a 3-year event window as the difference between the labor productivity of the combined firm one year after the transaction and the weighted average labor productivity of the acquiring and target firms one year before the acquisition, using annual employment shares as the weights. Our final sample consists of roughly 3,900 deals for which we observe information on productivity changes.

Because acquiring and target firms can operate in multiple industries, we compute a deal-level HCT index for each diversifying deal as a double-weighted average of the HCT index between each pair of industries operated by the acquiring and target firms as follows:

$$HCT_{A,T} = \sum_{A_i} w_{A_i} \sum_{T_j} w_{T_j} HCT(A_i, T_j).$$

For each acquirer-target industry pair, we first compute the weighted average HCT between each of the acquiring firm's industries and each target firm industry using the industry-level employment in the target firm as weights. Then, we compute the weighted average across all acquirer industries using industry employment levels in the acquiring firm as weights. Below, we often refer to the resultant measure as simply the HCT index for ease of exposition.

We estimate the following deal-level regression:

$$\Delta_{t-1,t+1} Y_i = \alpha + \beta_1 Divers_i + \beta_2 Divers_i \times High_HCT_{i,t-1} + X'_{it-1} \gamma + \varepsilon_{it}, \quad (2)$$

where $\Delta_{t-1,t+1} Y_i$ is difference between the natural logarithm of observed labor productivity of the merged firm in year $t+1$ and the natural logarithm of the weighted average of labor productivity across the target and acquirer firms in year $t-1$; *Divers* is an indicator variable that takes the value of one for diversifying deals; *High_HCT* is an indicator variable equal to one if the value of the HCT index is in the top quartile in our sample; and X is a vector of controls. It is not necessary to include both *High_HCT* and its interaction with *Divers* because *High_HCT* is only defined among diversifying deals. The variables *Divers* and *Divers* $\times$ *High_HCT* allow us to compare deals across three groups: within-industry deals (*Divers*=0 and *Divers* $\times$ *High_HCT*=0), low HCT diversifying deals (*Divers*=1 and *Divers* $\times$ *High_HCT*=0), and high HCT diversifying deals (*Divers*=1 and *Divers* $\times$ *High_HCT*=1). β_2 captures

labor productivity is highly correlated with total factor productivity in U.S. manufacturing firms. In [Internet Appendix Table OA9](#), we consider changes in total factor productivity on the subsample of manufacturing firms. We find broadly consistent evidence (TFP increases more among the target plants in high-HCT diversifying deals); however, our mechanism is likely to be strongest in (nonmanufacturing) firms that rely more on human capital in production.

Table 5
HCT Index and changes in labor productivity

Dependent variable	HCT using all workers		HCT using high-skill workers	
	Sales/Emp (1)	Sales/Payroll (2)	Sales/Emp (3)	Sales/Payroll (4)
DIVERS	-0.139 (0.164)	-0.113 (0.165)	-0.280* (0.162)	-0.231 (0.165)
DIVERS × High_HCT	0.184* (0.107)	0.159 (0.109)	0.299*** (0.105)	0.255** (0.110)
Overlap	-0.024 (0.075)	-0.022 (0.074)	-0.022 (0.075)	-0.02 (0.074)
Relative size	-0.047** (0.019)	-0.042** (0.018)	-0.048*** (0.019)	-0.042** (0.018)
ln(Employment)	-0.057*** (0.021)	-0.059*** (0.021)	-0.057*** (0.021)	-0.059*** (0.021)
Year Fixed Effects	Yes	Yes	Yes	Yes
Adjusted R ²	.05	.05	.05	.05
N (rounded)	3,900	3,900	3,900	3,900

This table presents estimates from OLS regressions of the change in labor productivity around acquisitions on deal characteristics. The dependent variable is the change in labor productivity for the combined firm around acquisitions, measured by the ratio of sales to employment (Sales/Emp) or sales to payroll (Sales/Pay). We use a 3-year event window surrounding each deal from $t - 1$ to $t + 1$ with year t being the transaction year. DIVERS is an indicator variable that equals one if the transaction is diversifying. High_HCT is an indicator variable that takes the value of one if the deal-level human capital transferability (HCT) index is in the top quartile among all transactions, based on all workers (columns 1 and 2) or workers with an annual wage greater than \$75K (columns 3 and 4). Overlap is the ratio of employment in industry groups operated by both the acquiring and the target firms to the total employment in the acquiring firm. Relative size is the log of the ratio of target's employment to acquirer's employment. ln(Employment) is the log of employment of the acquiring firm. Standard errors are clustered by year. The number of observations used in regressions is rounded to comply with the Census's disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

the difference in the productivity changes between deals involving industry combinations with high and low human capital transferability.

We include several additional controls in the analysis. We control for the degree of industry overlap between the acquirer and target firm (*Overlap*), which is the ratio of employment in industry groups operated by both the acquiring and target firms to the total employment in the acquiring firm. This variable allows us to compare the effect of human capital transferability on labor productivity to different benchmark groups of deals, depending on the degree of industry diversification.²² We add a control for the relative size of the acquiring and target firms (the natural logarithm of the ratio of target employment to acquirer employment) (*Relative Size*), and the size of the combined firm prior to the transaction (*ln(Employment)*). Because each deal appears only once in the sample, we need only adjust standard errors to account for heteroscedasticity.

We report the results using sales to employment as the labor productivity measure in column 1 of Table 5. Among the controls, we find that both *Relative size* and *ln(Employment)* have significant negative associations with productivity changes, consistent with higher costs of integrating new workers

22 The additional control does not have a material effect on the estimates of interest.

in larger deals. Turning to the coefficients of interest, diversifying deals with a low HCT index are associated with a relative decline in labor productivity, compared to within-industry deals, though the estimate is not statistically significant (coefficient for *Divers*). However, we find a positive and significant marginal effect of high human capital transferability: high-HCT deals have an 18.4-percentage-point larger increase in labor productivity than other diversifying deals. Taking together the coefficients for *Divers* and $Divers \times High_HCT$, we estimate similar productivity changes between high-HCT diversifying deals and within-industry deals.²³ It is important to note that the theory does not predict the relative size of the benefits from within-industry deals compared to high-HCT diversifying deals. In the former case, synergies come from economies of scale; whereas, in the latter case, they come from economies of scope. Thus, some benefits from diversifying deals—such as the collaboration of workers with different, but complementary skill sets—are not available in within-industry deals (and vice versa). However, the comparison still has value as a way to assess the economic magnitude of the synergies from human capital transferability. Overall, the results (1) confirm that human capital transferability is associated with greater value creation among diversifying deals and (2) suggest that the relative productivity gains are similar to the benefits from acquiring in the same product-market industry.²⁴

We also test whether the relative increase in sales among firms that diversify into high-HCT industries suffices to cover any associated increases in payroll. In column 2 of Table 5, we present the results of repeating the regression from column 1, but using sales to payroll as the productivity measure. Not surprisingly, the magnitude of the effects is smaller—workers with scarce skills can share in the rents from those skills (Lazear 2009)—and it is marginally insignificant.

In columns 3 and 4, we report the results from repeating the regressions in columns 1 and 2, but measuring *High_HCT* using the high-skill HCT index. We find that the magnitude of the coefficient for *High_HCT* is larger in both cases. The relative increase in productivity compared to low transferability diversifying deals is roughly 30% in column 3 and 25% in column 4. Both estimates are statistically significant. Thus, our results suggest greater benefits to diversifying deals when we focus on workers whose skills are likely to be scarcer in the external market.

²³ High-HCT diversifying deals have average productivity changes that are 4.5-percentage-points higher than nondiversifying deals, a difference that is not statistically significant ($0.045 = -0.139 + 0.184$; column 1, Table 5).

²⁴ We also consider how the effects evolve over time. We find that the negative effect of diversification disappears and becomes slightly positive (though still insignificant) if we compute the change in productivity through the end of the second full year following the deal. The estimated effect of high human capital transferability remains between 15 and 20%. Thus, over the extended horizon, high transferability diversifying deals appear to outperform within-industry deals. This makes sense if the presence of workers with related skills in other industries creates synergies and increases human capital investments that boost productivity relative to workers in stand-alone firms, as in Tate and Yang (2015).

As a robustness check, we reestimate the regressions using the median rather than the 75th percentile as the threshold for *High_HCT*, finding similar results. We also adjust for pre-trends in productivity by subtracting the weighted average growth rate in the acquiring and target industries, again with similar results. In [Internet Appendix Table OA8](#), we consider the extended set of controls from Section 3 (asset overlap, differences in Tobin's *q*, input/output relatedness) and deal-level analogs, such as the distance between acquirer and target establishments. We find similar results (the estimates of β_2 are slightly larger and are statistically significant in all specifications). We also consider a different, longer-term measure of deal performance: the hazard rate of subsequent divestiture of acquired industries. We find that divestiture rates over a 10-year postdeal period are significantly lower among high-HCT than among low-HCT diversifying deals ([Table OA10](#)).

4.2 Identification: Variation in the enforcement of noncompete agreements

Our estimates in Section 4.1 provide evidence of a robust positive correlation between human capital transferability among the industries operated by acquirers and targets in diversifying deals and postdeal performance. However, despite our efforts to control for the most obvious alternative channels that could preclude a causal interpretation of the evidence, it is still possible that an omitted factor correlated with the HCT index drives the positive association with performance. To make progress on this identification problem, we exploit plausibly exogenous variation in worker mobility across firms—and, therefore, the available benefits from internal labor markets—due to variation in the enforcement of noncompete agreements (NCAs) across states.

NCAs introduce constraints on employees' outside options. Stronger enforcement of NCAs decreases worker mobility in external labor markets ([Garmaise 2009](#); [Marx, Strumsky, and Fleming 2009](#)). In our model, then, NCAs provide plausibly exogenous variation in the adjustment costs that firms face in the external labor market.²⁵ Firms in states with stricter enforcement of NCAs enjoy larger net benefits from redeploying workers within an internal labor market, all else equal. Synergies from a high-HCT diversifying deal that are already positive in a weak enforcement state will be higher by the total amount of the expected adjustment cost savings in a strict enforcement state. For some low-HCT industry pairs, the adjustment cost savings from strict enforcement will turn negative synergy (forgone) diversifying deals into positive synergy deals. In these industry pairs, the frequency of deals will increase, but the positive synergies from the deal will be less than the

²⁵ Strict enforcement of noncompetes also reduces the (within-industry) outside options of a firm's workers, which could make them more willing to accept an internal redeployment.

full expected adjustment cost savings.²⁶ Within our empirical model from Section 4.1, then, we predict that stricter enforcement of NCAs leads to larger differences in the postdeal changes in productivity between deals in industry pairs with high and low HCT.²⁷

There is also a second potential channel. Starr (2019) shows that high NCA enforcement increases firm-provided training. Thus, a high enforcement regime not only reduces a firm's ability to hire talent externally but also can increase skill-development internally. If those acquired skills are transferable—which is more likely if the firm operates in high-HCT industry pairs—then strict enforcement can reduce the costs from redeploying workers internally to a second-best usage. Thus, this channel reinforces the prediction that postdeal changes in productivity are more positive among high-HCT deals in states where NCA enforcement is stricter.

We use the noncompete enforcement index from Starr (2019) for our analysis. Bishara (2011) assigns scores from 0 to 10 to each state on each of seven dimensions of NCA enforceability for the years 1991 and 2009.²⁸ Starr uses factor analysis on the scores to compute weights for the seven dimensions. The resultant (normalized) weighted sum is his state-level enforceability index (Starr 2019, table B1). Because our sample ends in 2009, we use the 1991 index. However, the correlation between the 1991 and 2009 indexes is 0.94, reflecting the stability of state policies over time. Following Starr (2019), we classify states in the top quartile to be high-enforceability states. Figure 3 presents a map of high- and low-enforceability states within the set of states in our Census sample. Notably, our sample includes the large states at the two extremes of the index (California, with virtual unenforceability, and Florida).²⁹ We define an acquiring firm to be in a high enforceability environment if the majority of its establishments are located in high-enforceability states.

Our identifying assumption is that the location of the acquirer's plants is predetermined and, therefore, plausibly exogenous to a deal. We do not directly exploit variation in the location of target firm establishments

²⁶ In industry pairs with very low HCT (e.g., agriculture and microchips), the costs from redeploying workers to jobs for which their skill sets are a poor match are likely to dominate adjustment cost savings, even where NCA enforcement is strict. Thus, these pairs will be irrelevant to our test, which conditions on the existence of a diversifying deal.

²⁷ We provide a more detailed development of the theory behind our empirical specification in the Internet Appendix.

²⁸ The seven dimensions capture (1) whether the state has a statute governing enforcement, (2) how broadly an employer's protected interest is defined, (3) how strong the employer's burden of proof is to establish existence of an enforceable NCA, (4) whether signing an NCA at employment inception suffices for enforceability, (5) whether continued employment suffices for enforceability of an NCA entered after the employment relationship has begun, (6) whether courts can modify overbroad NCAs, and (7) whether the NCA is enforceable if the employee is fired.

²⁹ Using a sample of SDC acquisitions to avoid the need for disclosure review, we find that 27% of deals among firms headquartered in our covered states are in high-enforceability states. The fraction is higher among diversifying deals (31%), consistent with heightened internal labor market synergies when NCAs create frictions in the labor market.

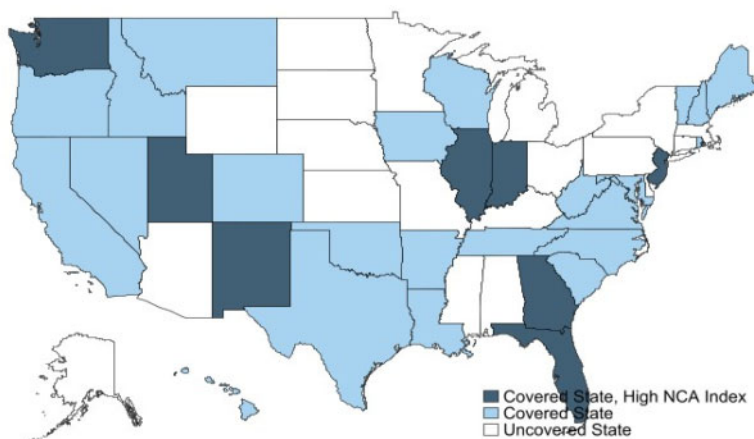


Figure 3

Noncompete states identified in our sample

The map shows states in our sample with the noncompete index (computed based on Starr 2019) in the top quartile.

because the acquirer chooses the location of the target.³⁰ We reestimate the regressions from Table 5, allowing the coefficient for the indicator for high-HCT diversifying deals to differ among acquirers in high- and low-enforceability regimes (and including the high-enforceability indicator and its interaction with the diversification indicator as controls).³¹ We report the results in Table 6. Using either measure of changes in labor productivity and either measure of human capital transferability, we find significantly larger increases in labor productivity in high-HCT deals among acquirers operating in high NCA enforceability regimes. By comparison, for acquirers operating in low enforceability regimes, we do not estimate a significant association between the HCT index and changes in labor productivity. We estimate a change in productivity following high-HCT diversifying deals that is 0.513 higher than among low-HCT deals in high enforceability regimes, but only 0.014 higher in low enforceability environments.³² Though not direct predictions of our internal labor market mechanism, the estimates also suggest that within-industry deals perform significantly better with strict noncompete

³⁰ Consistent with acquirers choosing targets in different locations if the constraint would bind, we do not find evidence that noncompete enforceability in target states matters for the effect of HCT on postdeal performance.

³¹ Because only diversifying deals can have a value of *High_HCT* equal to one, we cannot identify an interaction of *High_HCT* with the noncompete enforcement indicator given our other controls.

³² The estimated total effect is $0.129 \text{ (Divers)} + 0.014 \text{ (Divers} \times \text{High_HCT)} + 0.222 \text{ (Noncompete)} + 0.496 \text{ (Divers} \times \text{High_HCT} \times \text{Noncompete)} - 0.320 \text{ (Divers} \times \text{Noncompete)} = 0.541$ for high-HCT diversifying deals, and $0.129 + 0 + 0.222 + 0 - 0.320 = 0.031$ for low-HCT diversifying deals under strict NCA enforcement, which leads to a difference of $0.5417 - 0.0309 = 0.5108$. The estimated coefficient is $0.129 + 0.014 + 0 + 0 - 0 = 0.143$ for high-HCT diversifying deals, and $0.129 + 0 + 0 + 0 - 0 = 0.129$ for low-HCT diversifying deal under weak enforcement regimes, which leads to a difference of $0.143 - 0.129 = 0.014$.

Table 6
Changes in labor productivity: Noncompete agreement

Dependent variable	HCT using all workers		HCT using high-skill workers	
	Sales/Emp (1)	Sales/Payroll (2)	Sales/Emp (3)	Sales/Payroll (4)
DIVERS	0.129 (0.096)	0.122 (0.096)	0.098 (0.096)	0.095 (0.095)
DIVERS × High_HCT	0.014 (0.133)	−0.0004 (0.139)	0.152 (0.128)	0.119 (0.140)
DIVERS × High_HCT × Noncompete	0.496** (0.222)	0.466** (0.221)	0.438** (0.222)	0.406* (0.224)
Noncompete	0.222*** (0.055)	0.200*** (0.054)	0.221*** (0.055)	0.200*** (0.054)
DIVERS × Noncompete	−0.320** (0.155)	−0.289* (0.150)	−0.290** (0.152)	−0.268* (0.147)
Overlap	−0.019 (0.075)	−0.017 (0.074)	−0.018 (0.075)	−0.016 (0.074)
Relative Size	−0.047** (0.019)	−0.041** (0.018)	−0.048** (0.019)	−0.042** (0.018)
Ln(Employment)	−0.050** (0.021)	−0.053*** (0.021)	−0.051** (0.021)	−0.054*** (0.021)
Year fixed effects	Yes	Yes	Yes	Yes
Adjusted R-square	.05	.05	.05	.05
N (rounded)	3,900	3,900	3,900	3,900

This table presents estimates from OLS regressions of the change in labor productivity around acquisitions on the state-level enforceability of noncompete agreements. The dependent variable is the change in labor productivity for the combined firm around acquisitions, measured by the ratio of sales to employment (Sales/Emp) or sales to payroll (Sales/Pay). We use a 3-year event window surrounding each deal from $t-1$ to $t+1$ with year t being the transaction year. DIVERS is an indicator variable that equals one if the transaction is diversifying. High_HCT is an indicator variable that takes the value of one if the deal-level human capital transferability (HCT) index is in the top quartile among all transactions, based on all workers (columns 1 and 2) or workers with an annual wage greater than \$75K (column 3 and 4). Noncompete is an indicator variable that equals one if the majority of the acquiring firm's establishments are located in states in which the index of noncompete enforcement is in the top quartile following [Starr \(2019\)](#) and zero otherwise. Overlap is the ratio of employment in industry groups operated by both the acquiring and the target firms to the total employment in the acquiring firm. Relative size is the log of the ratio of target's employment to acquirer's employment. Ln(Employment) is the log of employment of the acquiring firm. Standard errors are clustered by deal. The number of observations used in regressions is rounded to comply with the Census's disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

enforcement (coefficient for *Noncompete*) and that low-HCT diversifying deals underperform within-industry deals in such environments (coefficient for *Divers × Noncompete*). A possible explanation is that some firms make within-industry acquisitions to relieve binding constraints on their ability to hire under strict NCA enforcement regimes.

Our NCA enforceability measure most directly captures exogenous variation in the costs to the acquirer to hire external talent in the local markets in which it operates. If our interpretation of the evidence is correct, then the firms that make acquisitions to expand the scope of their internal labor markets should choose targets in closer geographic proximity to the acquirer's existing establishments. Because workers are free to leave the firm, the firm's cost to reallocate workers between acquirer and target establishments is also likely to depend on workers' relocation costs, which are smaller when the establishments are close together. Moreover, if the target establishments are in the same legal jurisdiction, strict NCA enforceability also means that the

target workers who are newly part of the firm are less likely to reject future redeployment because they have fewer local external job opportunities. To test this prediction, we compute the employment-weighted average of the distance between establishments of the acquiring and target firms in diversifying deals. We find that the distance among high-HCT deals is roughly 100 miles smaller than among low-HCT deals, a difference that is significant at the 10% level (the difference to within-industry deals is roughly 60 miles).³³ Similarly, the fraction of deals with acquirer and target establishments within 100 miles of each other is roughly double among high-HCT deals compared to low-HCT or within-industry deals (differences that are significant at the 1% level).

4.3 Alternative performance measure: Stock market reactions

Our analysis thus far identifies acquisitions using changes in firm identifiers within the Census' administrative databases. Because the sample contains deals among both privately held and publicly traded firms, we cannot use stock prices to compute measures of (perceived) deal synergies. Here, we consider an alternative sample of deals between publicly traded firms to test whether our conclusions hold using standard return-based measures of deal performance.

We consider the sample of completed acquisitions from the SDC's U.S. Mergers and Acquisitions Database for which the initial bid was announced between January 1, 1994, and December 31, 2008. We obtain the industry compositions of the acquirer and target from the Computat segment file. In [Internet Appendix Table OA11](#), we show a positive association between the HCT index and industry choices in diversifying deals on this subsample of deals, confirming comparability to the overall sample. We use the stock price information that is available for these deals to measure differences in value creation. We find that the combined CARs at the announcement are roughly 90 basis points higher among high-HCT than among low-HCT deals. The effect is statistically significant at the 10% level. We find stronger effects when we use the high-skill HCT measure (the magnitude of the difference is nearly double). We report the results in [Table 7](#). Economically, the differences are large enough to fully offset the negative market reaction to diversification. We also consider announcement returns to the acquiring and target firms separately. In general, we find economically larger effects on target returns, but the relation between low- and high-HCT deals is similar to what we observe in combined returns.

It is important to note that stock price reactions do not measure future firm profitability directly, but instead measure the market's beliefs about future value creation. Moreover, the beliefs are about all expected changes in future productivity, not just short-run productivity changes. Nevertheless, results from the SDC sample are consistent with our inferences from [Table 5](#), in which

³³ The average employment-weighted distance between the acquirer and the target firms are 753, 800, and 692 miles for within-industry, low-HCT, and high-HCT deals, respectively.

Table 7
HCT Index and announcement returns (SDC sample)

Dependent variable	HCT using all workers			HCT using high-skill workers		
	CCAR(−1,1) (1)	TCAR(−1,1) (2)	ACAR(−1,1) (3)	CCAR(−1,1) (4)	TCAR(−1,1) (5)	ACAR(−1,1) (6)
DIVERS	−0.802* (0.447)	−2.281*** (0.842)	−0.273 (0.484)	−0.911* (0.497)	−2.213** (0.916)	−0.290 (0.510)
DIVERS × High_HCT	0.897* (0.472)	3.145** (1.499)	0.260 (0.593)	1.517** (0.593)	3.196* (1.637)	0.373 (0.728)
D_Stock	−1.567*** (0.437)	−4.362*** (1.069)	−1.232*** (0.455)	−1.572*** (0.435)	−4.342*** (1.083)	−1.232*** (0.454)
Relative size	1.016*** (0.292)	−0.800 (0.557)	−0.125 (0.312)	1.011*** (0.289)	−0.811 (0.562)	−0.126 (0.311)
IO_Related	0.439 (0.577)	2.166 (1.379)	0.191 (0.630)	0.181 (0.554)	1.940 (1.312)	0.138 (0.650)
ln(Asset)	−0.413*** (0.086)	0.827** (0.313)	−0.316** (0.122)	−0.416*** (0.088)	0.825** (0.314)	−0.317** (0.123)
Deal Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-square	.08	.07	.03	.08	.07	.03
N (rounded)	4,300	4,300	4,300	4,300	4,300	4,300

This table presents regression estimates of deal outcome measures on human capital transferability. We use deals in which both acquirer and target firms are publicly traded, and the dependent variables are combined cumulative abnormal returns (CCAR), target cumulative abnormal returns (TCAR), and acquirer cumulative abnormal returns (ACAR), respectively, all expressed as percentages. We calculate market-adjusted abnormal returns using the CRSP equal-weighted index to measure market returns. CCAR is the weighted average of TCAR and ACAR using market values 50 days prior to the announcement as weights. DIVERS is an indicator variable that equals one if the transaction is diversifying. High_HCT is an indicator variable that takes the value of one if the value of the human capital transferability (HCT) index based on all workers or workers with annual wages greater than \$75K is in the top quartile among all transactions, computed one year before the transaction. D_Stock is an indicator variable equal to one if more than 50% of the transaction is financed by stock. Relative size is the natural logarithm of the ratio of the target's market value over the acquirer's market value. IO_Related is an indicator variable equal to one if input/output flows between the industries in which the acquirer and target firms operate are greater than 2.5% of total industry output. ln(Asset) is the logarithm of the acquirer's total assets. Standard errors are robust to heteroscedasticity. The number of observations used in regressions is rounded to comply with the Census's disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

we measure changes in labor productivity directly over the 2 years following a deal.

5. Prediction 4: Human Capital Transferability and Worker Outcomes

Our model also makes prediction for postmerger outcomes at the worker level. If firms pursue diversification to enrich internal labor markets, then we expect to see evidence of higher retention rates, particularly among high-skill workers, and more frequent internal cross-industry job changes (Prediction 4). The productivity gains that we measure in Section 4 can operate through mechanisms other than worker retention and reallocation decisions, such as enhanced worker collaboration and ongoing human capital investments. We cannot measure these channels directly in the data; nevertheless, they are likely to be correlated with differences in job changes. Thus, we use job changes as a proxy for the larger set of worker-level outcomes.

To track employment status for workers in target firms following an acquisition, we combine our acquisition sample with worker-level information

from the LEHD data.³⁴ We include workers employed by the target firm two quarters prior to the acquisition and require that they have worked at the target firm for at least one year. Though the merged firm could theoretically benefit in the short term, either moving workers from the target to the acquirer or vice versa, we focus on target workers because all target workers experience a change in management following the deal, and thus face more uncertainty in terms of retention and redeployment.

5.1 Worker retention

First, we test whether acquirers in high-HCT deals are more likely to retain skilled workers from the target firm following a deal than acquirers in low-HCT or within-industry deals. Our sample contains about 1.4 million workers from 3,700 transactions.

We estimate the following linear probability model on the full sample of workers employed by target firms two quarters prior to each deal in the sample:

$$\begin{aligned} \text{Retain}_{ij} = & \alpha + \beta_1(\text{Divers}_i \times \text{Low HCT}_{i,t-1}) + \beta_2(\text{Divers}_i \times \text{High HCT}_{i,t-1}) \\ & + \beta_3 \text{Skill}_j + \beta_4(\text{Skill}_j \times \text{Divers}_i \times \text{Low HCT}_{i,t-1}) \\ & + \beta_5(\text{Skill}_j \times \text{Divers}_i \times \text{High HCT}_{i,t-1}) + \mathbf{X}'_i \boldsymbol{\gamma} + \boldsymbol{\varepsilon}_i \end{aligned} \quad (3)$$

for worker j in target firm i . *Retain* is an indicator variable equal to one if the worker remains employed by the merged firm four quarters following the deal. *Divers* is an indicator variable equal to one if the acquirer and target do not operate in any common Fama-French industry groups prior to the acquisition. *Low HCT* and *High HCT* are indicator variables that equal one if the value of the HCT index measured one year prior to the deal is below or above the median, respectively. Since HCT is only specified for diversifying deals, we can define three groups of acquisitions using the two interactions: within-industry acquisitions ($\text{Divers} \times \text{Low HCT} = 0$ and $\text{Divers} \times \text{High HCT} = 0$), diversifying acquisitions with low-HCT ($\text{Divers} \times \text{Low HCT} = 1$ and $\text{Divers} \times \text{High HCT} = 0$), and diversifying acquisitions with high-HCT ($\text{Divers} \times \text{Low HCT} = 0$ and $\text{Divers} \times \text{High HCT} = 1$). *Skill* is a proxy for worker skill level, and $\mathbf{X}$ is a vector of controls. In the vector of controls $\mathbf{X}$, we include standard demographic controls: the natural logarithm of worker age and indicator variables for female workers, managers (measured as the highest-paid worker in the reporting unit), and six race categories (with white workers as the omitted baseline). We include the natural logarithm of worker tenure and the natural logarithm of the worker's annual wage, both of which are measured before the deal. We also include a number of firm- and deal-level controls. First, we include (separately) the natural logarithms of employment in the acquiring and target firms prior to the deal as a rough control for the availability of internal opportunities. Likewise,

³⁴ We exclude deals in which either the target or acquiring firms have establishments outside the 31 LEHD states.

we include the number of different Fama-French industry groups in which the acquirer already operates prior to the deal and an indicator variable that equals 1 if the acquirer also operates in the state in which the target worker is employed prior to the deal. In addition, we include two controls to capture differences in the outcomes of workers from target firms that were already undergoing restructuring prior to the deal in our sample: the changes in the number of plants operated by the firm and total employment in the firm in the year prior to the acquisition.³⁵ Finally, we include deal year, state, and target firm industry fixed effects to capture differences in market conditions across years, states, and industries. We estimate a linear specification to ease the interpretation of the coefficients for interaction terms.³⁶ We cluster standard errors at the deal level to correct for correlation of the residuals among workers who are part of the same deal.

In Table 8, we present the results. We use an indicator for workers with annual wages higher than the within-firm 75th percentile as the measure of worker skill (*Skill*).³⁷ In column 1, we use the HCT index constructed from the full sample of job changers to define *High HCT* and *Low HCT*. We find that the marginal effect of high human capital transferability on the likelihood of retaining high-skilled workers is positive and significant (the coefficient for $Skill \times Divers \times High\ HCT = 0.051$). In column 2, we instead define *High HCT* using the high-skill HCT index. The results are similar. Translating the marginal effects to differences in retention rates across acquisition types using low-skill workers in within-industry deals as the benchmark, the relative retention rate of high-skill workers is -4% among within-industry deals and -2.9% among low-HCT diversifying deals, but is 2.2% among high-HCT diversifying deals.³⁸ We see much smaller relative differences in the retention of low-skill workers (the relative rates are 1.9% and 0.8% among low- and high-HCT diversifying deals, respectively). Among the controls, we find, not surprisingly, a general tendency toward retaining higher-wage workers. However, managers are unlikely to be retained.³⁹ Also, workers with more experience (age, tenure) are more likely to remain with the merged firm. Our estimates of the relation between HCT and retention are similar if we estimate the regressions only on the subsample of high-skill workers to address the

³⁵ These controls are potentially important for the worker-level tests because it is more difficult to accurately link workers to firms in cases of successive restructuring. We observe ownership changes in the LBD at an annual frequency, but worker information from the LEHD data at a quarterly frequency.

³⁶ Here, and throughout the worker-level analysis, the results are similar if we estimate logit specifications.

³⁷ The results are not sensitive to using this cutoff and, for example, are similar using the median.

³⁸ Using estimated coefficients from Table 11, column 2, the relative retention rate is $-4\% + 1.9\% - 0.8\% = -2.9\%$ for high-skill workers in low-HCT diversifying deals, and $-4\% + 0.8\% + 5.4\% = 2.2\%$ for high-skill workers in high-HCT diversifying deals.

³⁹ This result also confirms that our mechanism is distinct from the effects of managerial background explored by Custodio and Metzger (2014).

Table 8
HCT Index and worker retention

Dependent variable D_Retention	HCT (1)	High-skill HCT (2)
Divers × Low_HCT	0.032 (0.020)	0.019 (0.020)
Divers × High_HCT	−0.002 (0.025)	0.008 (0.025)
Skill	−0.040*** (0.006)	−0.040*** (0.006)
Skill × Divers × Low_HCT	−0.004 (0.024)	−0.008 (0.024)
Skill × Divers × High_HCT	0.051** (0.024)	0.054** (0.023)
Female	0.021*** (0.004)	0.021*** (0.004)
Manager	−0.179*** (0.014)	−0.179*** (0.014)
ln(Age)	0.139*** (0.010)	0.139*** (0.010)
ln(Wage)	0.106*** (0.007)	0.106*** (0.007)
ln(Tenure)	0.093*** (0.004)	0.093*** (0.004)
ln(Target_Emp)	0.012*** (0.004)	0.012*** (0.004)
ln(Acq_Emp)	−0.004 (0.004)	−0.003 (0.004)
Chg(N_Estabs)	0.039 (0.038)	0.040 (0.038)
Chg(FirmEmp)	−0.039 (0.024)	−0.038 (0.024)
Acq_#ofInds	0.006*** (0.002)	0.006*** (0.002)
Same_State	−0.027* (0.016)	−0.027* (0.016)
Year fixed effects	Yes	Yes
State fixed effects	Yes	Yes
Target industry fixed effects	Yes	Yes
Adjusted R-square	.14	.14
N (rounded)	1,400,700	1,400,700

This table presents, estimates from linear probability models on the sample of workers employed by target firms before acquisitions. The dependent variable equals one if the worker is retained after the transaction and zero otherwise. We include all target firm workers. We define skilled workers (Skill) as workers whose wage is in the top quartile in the firm. Divers is an indicator variable that equals one for diversifying deals and zero otherwise. Low_HCT and High_HCT are indicator variables that equal one if the deal-level HCT index is below or above the median in our sample, respectively. Female is an indicator for female workers, and Manager is an indicator for the highest-paid employee in the firm. ln(Wage) and ln(Tenure) are the log of worker wage and worker tenure, respectively. ln(Target_Emp) and ln(Acq_Emp) are the log of employment in the target and acquiring firm, respectively. Chg(N_Estabs) and Chg(FirmEmp) measure the change in the number of establishments and the total employment in the target firm one year before the acquisition. Acq_#ofInds measures the number of industries operated by the acquiring firm prior to the acquisition. Same_State is an indicator that equals one if the acquiring and the target firm operate in the same state. We include six indicator variables for race categories (coefficient estimates not reported). Standard errors are clustered at the firm level and are reported in parentheses. The number of observations used in regressions are rounded to comply with the Census’s disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

possibility that the controls have different effects among low- and high-skill workers.

As a robustness check, we rerun the regressions using long tenure with the firm—specifically, an indicator of tenure longer than 5 years—as an alternative

proxy for skill. Workers with longer tenure are likely to be higher skilled both because of selection (the firm will prefer to retain workers who prove to be highly skilled after information about worker type is revealed) and because of accumulated firm-specific human capital. We find qualitatively similar results: high HCT is associated with a higher likelihood of retaining workers with longer tenure. We also consider the likelihood of remaining in the firm eight quarters following the deal. We again find similar patterns: acquiring firms that diversify into high HCT industries with respect to their existing operations are more likely to retain high-skilled workers, even 2 years following the deal.

A potential alternative explanation of the results is that within-industry and low-HCT acquirers engage in more successful cost-cutting following acquisitions by trimming experienced workers with bloated salaries. If this were the case, we should expect such deals to improve labor productivity relative to high-HCT diversifying deals. Yet, we find the opposite. Labor productivity increases most in high-HCT deals, even measured relative to payroll. Our story does not rest on the assumption that firms involved in either type of deal are operating inefficiently, before or after the deal. Diversifying deals with high human capital transferability simply have additional sources of synergies, including the real option for the acquirer to reallocate human capital.

5.2 Worker internal movement

Next, we test whether high-HCT acquirers reallocate human capital from the target elsewhere in the merged firm. These tests provide direct worker-level evidence for the internal labor markets mechanism. Because we are interested in industry changes within internal labor markets, we only consider the subset of diversifying deals, and compare deals with different levels of human capital transferability. We estimate the following specification:

$$Move_{ij} = \alpha + \beta \times High_HCT_{i,t-1} + X'_{ij} \gamma + \varepsilon_i, \quad (4)$$

for worker j in deal i . *Move* is an indicator variable equal to one if the worker moved to one of the acquirer's predeal reporting units after the acquisition, thereby changing industries. We measure *Move* four or eight quarters after the transaction, conditional on retention. *High_HCT* is an indicator variable equal to one if the value of the HCT index measured one year prior to the deal is above the median, and X is a vector of controls.⁴⁰ We include the same controls in X as we did in our estimation of worker retention probabilities in Table 8, including deal year, state, and target industry fixed effects. We also again report estimates of a linear regression model—here because we include multiple fixed

⁴⁰ Because the sample consists of diversifying deals only (it is impossible to move to a new industry inside the merged firm following a focused deal), it is not necessary to consider a level effect of diversification.

Table 9
HCT Index and worker movement

	A		B	
	HCT using all workers		HCT using high-skill workers	
	<i>t</i> + 4 (1)	<i>t</i> + 8 (2)	<i>t</i> + 4 (3)	<i>t</i> + 8 (4)
High HCT	0.101*** (0.031)	0.123*** (0.039)	0.097*** (0.033)	0.128*** (0.039)
Female	0.006 (0.010)	0.004 (0.012)	0.006 (0.010)	0.005 (0.012)
Manager	0.003 (0.017)	−0.005 (0.026)	0.007 (0.018)	0.001 (0.026)
ln(Age)	−0.010 (0.008)	−0.001 (0.011)	−0.008 (0.008)	0.002 (0.011)
ln(Wage)	0.000 (0.008)	0.003 (0.011)	−0.001 (0.008)	0.002 (0.011)
ln(Tenure)	0.008 (0.007)	0.009 (0.009)	0.009 (0.007)	0.009 (0.009)
ln(Target_Emp)	−0.035** (0.014)	−0.050** (0.020)	−0.034** (0.014)	−0.048** (0.020)
Ln(Acq_Emp)	−0.003 (0.009)	−0.014 (0.014)	−0.005 (0.009)	−0.018 (0.014)
Chg(N_Estabs)	−0.160* (0.084)	−0.168** (0.080)	−0.165* (0.086)	−0.179 (0.083)
Chg(FirmEmp)	−0.029 (0.039)	−0.028 (0.045)	−0.012 (0.038)	−0.005 (0.045)
Acq_#ofInds	−0.003 (0.006)	−0.007 (0.009)	−0.001 (0.006)	−0.005 (0.009)
Same_State	0.007 (0.031)	0.017 (0.040)	0.010 (0.031)	0.021 (0.041)
Year fixed effects	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes
Target industry fixed effects	Yes	Yes	Yes	Yes
Adjusted R-square	.35	.33	.35	.33
N (rounded)	67,200	53,700	67,200	53,700

This table presents estimates from linear probability models on the sample of workers from target firms who were retained following diversifying acquisitions. The dependent variable equals one if the worker moves to a new industry within the merged firm and zero otherwise. Columns 1 and 2 use the human capital transferability (HCT) index based on all workers, and columns 3 and 4 use the HCT index based on workers with wages greater than \$75K. We measure worker outcomes four quarters (columns 1 and 3) and eight quarters (columns 2 and 4) after the transaction. High_HCT is an indicator variable that equals one if the deal-level HCT index is above the median in our sample. Female is an indicator for female workers, and Manager is an indicator for the highest-paid employee in the firm. ln(Age), ln(Wage) and ln(Tenure) are the log of worker age, wage, and tenure, respectively. ln(Target_Emp) and ln(Acq_Emp) are the log of employment in the target and acquiring firm, respectively. Chg(N_Estabs) and Chg(FirmEmp) measure the change of the number of establishments and the total employment in the target firm one year before the acquisition. Acq_#ofInds measures the number of industries operated by the acquiring firm prior to the acquisition. Same_State is an indicator that equals one if the acquiring and the target firm operate in the same state. We include six indicator variables for race categories (coefficient estimates not reported). Standard errors are clustered at the firm level and are reported in parentheses. The number of observations used in regressions are rounded to comply with the the Census’s disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

effects, which can bias coefficient estimates in a nonlinear model—and cluster standard errors at the deal level.

We report the results of estimating Equation (4) in Table 9. In panel A, we use the HCT index based on all workers’ job changes to define *High_HCT*. In column 1, the dependent variable *Move* is defined by observing each target firm worker’s employing unit four quarters following the acquisition. We find

few robust predictors of internal industry changes among the control variables. An exception is the total number of employees in the target firm prior to the deal, which has a negative effect on the probability of changing jobs in the internal market. The strongest predictor is the indicator for high-HCT deals. We find a 10.1-percentage-point increment to the estimated probability of changing industries, an effect that is statistically significant at the 1% level. The average frequency with which workers change industries in the sample is 7%, suggesting that the estimated effect is also economically large. In column 2, we redefine the dependent variable *Move* by instead considering the workers' employing units eight quarters following the transaction. We find a stronger effect. Here, the probability of moving to a job in a new industry in the acquiring firm is roughly 12 percentage points when HCT is high.

In panel B, we report the results using the high-skill HCT index to define *High_HCT*. We report the results from estimating the probability of moving to the acquirer four (column 3) and eight (column 4) quarters following the deal. The estimates are not materially different from those we report in panel A. Again, the workers from the targets in high-HCT deals are more likely to move to the acquirer's units following the merger. In untabulated regressions, we also include interactions of the HCT variable with proxies for worker skill (high wages or long tenure). We do not find that the probability of moving to the acquirer differs for low and high-skill workers in high-HCT transactions using either measure. While the benefits to the firm from the mobility of high-skill workers may be larger—explaining the larger productivity differences and predictive power for the decision to merge—these results suggest that the firm exercises (and likely profits from) the ability to redeploy workers of all types. As a final step, we test whether the target workers who move to jobs in acquirer industry segments indeed move to the industries that rank highest by the HCT index with their industry of origin. Consistent with our story, we find that 82% of workers move to a new industry that is in the upper half of the acquirer's industries by HCT with their original industry (84% if we use the high-wage version of the index). Taken together with the productivity results from Section 4, these findings suggest that the high-HCT segments to which workers move are also likely to have high expected productivity.

Overall, we find evidence at the worker level for the internal labor markets mechanism that we propose as a motivation for some diversifying deals. When there is high transferability of human capital between the industries operated by merging firms, the merged firm is better able to retain high-skill workers following the deal. Moreover, the firm is more likely to reallocate the acquired workers to its own units operating in different industries and, most often, to the ones to which target worker skills have high transferability. Importantly the short-run transfers that we observe in the year following a deal are unlikely to be the sole source of expected value from the internal market; instead, firms

can expect similar transfers in both directions in response to future productivity shocks.

6. Human Capital Transferability and Firm Composition

The main objective of our study is to investigate the role human capital transferability plays in explaining the decisions of firms to expand the scope of their operations. As a result, we consider event studies around mergers, comparing the outcomes of deals with high and low human capital transferability between the acquirers and targets. As a final step, however, we abstract from merger events. We test whether the industry composition of diversified firms in the cross-section correlates with our measures of human capital transferability. This test provides one way to evaluate the economic importance of our mechanism. If human capital transferability is a first-order consideration, then it should show up as a predictor of firm composition.

To conduct our test, we reconsider Equation (1) from Section 3. However, instead of defining the dependent variable using diversifying acquisitions within industry pairs, we instead measure the frequency with which each industry pair is jointly operated by existing diversified firms. For each diversified firm in the LBD, year by year, we identify industry segments that make up at least 10% of total employment. Then, for each Fama-French industry, we define the dependent variable as the fraction of diversified firms operating in that industry that also operate a segment in each of the other distinct Fama-French industries. Thus, the unit of observation is again an industry-pair year. We include the percentage of output that flows between the two industries, measured using the input-output matrix, to control for the relatedness of the products produced by the industries. We also include industry by year fixed effects for each industry in a pair and double cluster the standard errors by the two industries.⁴¹

We report the results in Table 10. In column 1, we use the HCT index defined using all workers' job changes over a 5-year rolling window ending one year prior to the year in which we measure the industry composition of the diversified firms. We find a strong and significant positive relation between human capital transferability and the industries that are jointly operated by diversified firms. As in Section 3, we find that human capital transferability has more explanatory power for the composition of diversified firms than the product market relatedness of the industries. The coefficient estimate on the HCT index is more than twenty times the size of the estimate on input-output relatedness, even though a standard deviation of HCT is roughly 40% of a

⁴¹ We find consistent results if we instead include industry-pair fixed effects and identify the estimates out of the time series, though with smaller economic magnitudes (see Internet Appendix Table OA13). However, as in Section 3, the bulk of the variation in our measure comes from the cross-section.

Table 10
HCT Index and industry portfolio in diversified firms

	HCT using all workers (1)	HCT using high-skill workers (3)
HCT Index	2.080*** (0.317)	1.877*** (0.353)
Pct_Related	0.135*** (0.044)	0.110*** (0.047)
Ind1-year fixed effects	Yes	Yes
Ind2-year fixed effects	Yes	Yes
Adjusted R-square	.53	.52
N (rounded)	27,800	27,800

This table reports coefficient estimates from OLS estimation of equation (1) on the sample of industries operated by existing diversified firms. The sample contains all industry pairs by year. The dependent variable is the intensity with which a firm operating in industry i also operates in industry j (for $i \neq j$) in year t . HCT Index is the human capital transferability index between industry i and j measured over a five-year window ending at $t - 1$. We use the HCT index based on all workers in column 1 and the HCT index based on workers with wages greater than \$75K in column 2. Pct_Related measures the linkage through I/O markets between the industry pair. Standard errors are double-clustered by industry. The number of observations used in regressions is rounded to comply with the Census’s disclosure policy. * $p < .1$; ** $p < .05$; *** $p < .01$.

standard deviation of input-output relatedness. In column 2, we repeat the estimation, but using the high-skill HCT index. The results are similar.

An immediate concern in interpreting these results could be that the job changes on which we base our measure of human capital transferability are simply job changes within diversified firms in the sample, inducing reverse causality. However, recall that we use only job changes in which workers exit their original employing firm to construct our indexes, ruling out this possibility. Moreover, we consider only job changes in which some, but not all, of the workers from a firm change jobs in a given quarter. This restriction prevents changes in firm identifiers due to mergers from contaminating our sample of job changes. Thus, our tests indeed isolate a tendency for diversified firms to operate in sets of industries between which worker skills easily transfer. Moreover, the existence of the association between this transferability and firm composition in the cross-section—and its relative strength compared to product market links—suggests that human capital transferability is a first-order determinant of firm scope.

7. Conclusion

We study the transferability of human capital as a motivation for diversifying acquisitions. We use worker-firm matched data from 31 states available from the LEHD program to construct a measure of this transferability using the frequency of worker job changes between industries. We find that human capital transferability not only predicts the intensity with which firms diversify into other industries but also predicts the intensity more strongly than measures of product market relatedness. Moreover, it predicts the industry composition of diversified firms in the cross-section.

We also find that firms reap tangible benefits from diversifying into high HCT industries with respect to their existing divisions. Firms that acquire industries with greater human capital transferability with their existing industries enjoy larger increases in labor productivity and are less likely to divest their newly acquired industries subsequently.

Finally, we find evidence consistent with the human capital channel at the worker level. Acquiring firms that diversify into industries with high human capital transferability with their existing operations are more likely to retain skilled workers following the deal and to transfer workers from the target firm to other industries in the merged firm.

Our results challenge the view that diversification across unrelated product markets destroys value or is a manifestation of agency problems. Instead, we identify potential synergies from diversification based on the transferability of human capital across industries and the benefits of internal labor markets. Moreover, our theory does not require that acquirer, target, or merged firms are operating inefficiently to justify the transactions. Thus, it is likely to be a mistake to equate all diversification with a failure of governance. Though we do not claim that all diversifying deals create value or that there is no role for agency models, our results provide one explanation for why many large, successful firms in U.S. markets are diversified. An interesting avenue for additional study would be to link general patterns in labor mobility with trends in corporate restructuring activity over longer time horizons.

Our results also have implications for how we think about firm groupings, like industries, in finance and economics. Traditionally, researchers have focused exclusively on product market considerations to group firms. An industry consists of firms that sell similar products. Similarly, industries are related if the products produced by one industry are used as inputs for the other. We instead identify human capital as an important common factor across firms. Recent asset pricing research argues that this commonality is a risk factor in equity markets (Eisfeldt and Papanikolaou 2013; Donangelo 2014). We argue that human capital transferability is also an important determinant of firm boundaries. Another opportunity for future research is to build on this evidence, considering the implications of human capital shocks for corporate outcomes in much the same way traditional corporate finance considers product market industry shocks.

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